

Section I. GENERAL INFORMATION

1-1. GENERAL. The C-130 Lighting System fault analysis concept is a troubleshooting method composed of two elements: the Fault Reporting Manual (FRM) and the Fault Isolation Manual (FIM).

NOTE

Ground maintenance crews may also use FRM procedures (contained in Section XIV of TO 1C-130H-2-00FR-00-1) to gain the same results as described in the following paragraph.

1-2. FAULT REPORTING MANUAL (FRM). Flight crews use FRM methods to inform maintenance personnel about malfunctions the flight crews encounter. As directed by the FRM procedures, the flight crew makes checks and observations to help isolate the malfunction. Questions that deal with symptoms and conditions occurring at the time of the failure are answered. The result is a peculiar fault code which represents those specific symptoms and conditions accompanying the malfunction. In order that each fault code may be identified and continuity made possible between the fault reporting manual (FRM) and the fault isolation manual (FIM), a seven-digit numeric code is assigned to each fault. The first two digits identify the system in which the fault occurred; the next two digits represent the subsystem in which that fault occurred; the remaining three digits are the numerical identification assigned to each fault. System and subsystem identification are assigned in accordance with Specification MIL-M-83495. This number and the discrepancy (symptoms) are then entered in the AFTO Form 781A along with any additional details that will be helpful. Using the Fault Isolation Manual, maintenance personnel interpret the fault code and plan maintenance action that allows them to start the fault isolation routine at some block other than block 1. The blocks which are skipped over would normally be used to determine the symptoms and conditions represented by the fault code. Troubleshooting becomes more accurate, and less time and effort is required.

1-3. FAULT ISOLATION MANUAL (FIM). Maintenance personnel use FIM procedures to isolate faults for a system, a functional mode, or a specific failure indication. The procedures are based on a process of elimination; the parts of a system/malfunction are tested in a logical sequence until a specific line replaceable unit (LRU) failure or a wiring fault is found. This eliminates the guesswork replacement of components which are operating properly.

a. Contents. Contents of each section of the FIM are:

- (1) Malfunction indexes.
- (2) Fault code indexes.
- (3) Preparation procedures to be performed prior to fault isolation.
- (4) Fault isolation diagrams (the actual troubleshooting procedures).
- (5) Locator illustrations for hardware referenced in the fault isolation diagrams.
- (6) Schematic diagrams.

b. Malfunction Indexes. The malfunction indexes list malfunctions according to the system and/or mode of operation when the failure occurred, and also contain the alphanumeric fault code. These indexes are to be used when the malfunction occurs during ground maintenance or when the flight crew did not provide a numeric fault code. When using these indexes, troubleshooting usually begins at block 1 of the specified fault isolation diagram.

c. Fault Code Indexes. The fault code indexes list all applicable numeric fault codes in numerical order. Each fault code represents a peculiar malfunction and its associated symptoms